GABRIEL S. STUART

Automation & Computer Vision Developer · Systems Simulation

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Builds full computer-vision and simulation systems end-to-end — from automated chart-data extraction pipelines to industrial plant simulators and multi-agent digital twins, in Python, C/C++, and Java.

KEY SKILLS

Languages: Python, C/C++, Java

Computer Vision: OpenCV, YOLO, ONNX, OCR, chart/PDF data extraction

Simulation: event-driven architecture, thermodynamic modeling, gRPC co-simulation

Multi-Agent Systems: JaCaMo/Jason, PROSA/ADACOR, Contract-Net negotiation, 2PC

Engineering & DevOps: pytest/JUnit, GitHub Actions CI/CD, Docker, Numba/Cython

Data & Statistics: Cohen's Kappa, Lin's CCC, hypothesis testing, OPEX/LCOH analysis

FEATURED PROJECTS

Plot-in *github.com/utczbr/Plot-in · Python*

- Computer-vision pipeline extracting quantitative data from charts and PDFs across 8 chart types (bar, line, scatter, box, histogram, heatmap, pie, area).
- Statistical validation framework (success rate, Lin's CCC, Cohen's Kappa); cross-platform installers and CI/CD via GitHub Actions.

h2_plant — Hydrogen Production Plant Simulator *github.com/utczbr/h2_plant · Python*

- Modular six-layer simulator covering electrolysis and ATR routes, with an event-driven engine and checkpointing.
- JIT-optimized (Numba) thermodynamic property calculations, 95%+ test coverage, and a proprietary OPEX/LCOH techno-economic model.

Matrix_Factory *github.com/utczbr/Matrix_Factory · Python, Java*

- Co-simulation digital twin of a PEM fuel-cell line, coupling a Python physics engine with a Java (JaCaMo) multi-agent layer via gRPC.
- PROSA/ADACOR holonic agents with Contract-Net negotiation; Cooperative A* AMR fleet routing; automated pytest + JUnit test suite.

PROFESSIONAL EXPERIENCE

External Developer — ROGER Renewable Energy 11/2025 – 05/2026

- Developed dynamic physical simulation for a hydrogen plant; led CAPEX/OPEX validation.

Builder — Robota Competition Team 7/2022 – 9/2023

- Developed line-following robots.

EDUCATION

UFSC (Federal University of Santa Catarina) — Control and Automation Engineering Expected 2027

Graduate coursework (non-degree, PPGEAS/UFSC, 2024–2026): AI, Concurrency & Real-Time Systems, Renewable Energy Systems, Machine Learning, Mobile Robotics, Linear Dynamic Systems, Multiagent Systens.

LANGUAGES

Portuguese — Native English — Advanced